Parallel and Distributed Computing

Assignment 2

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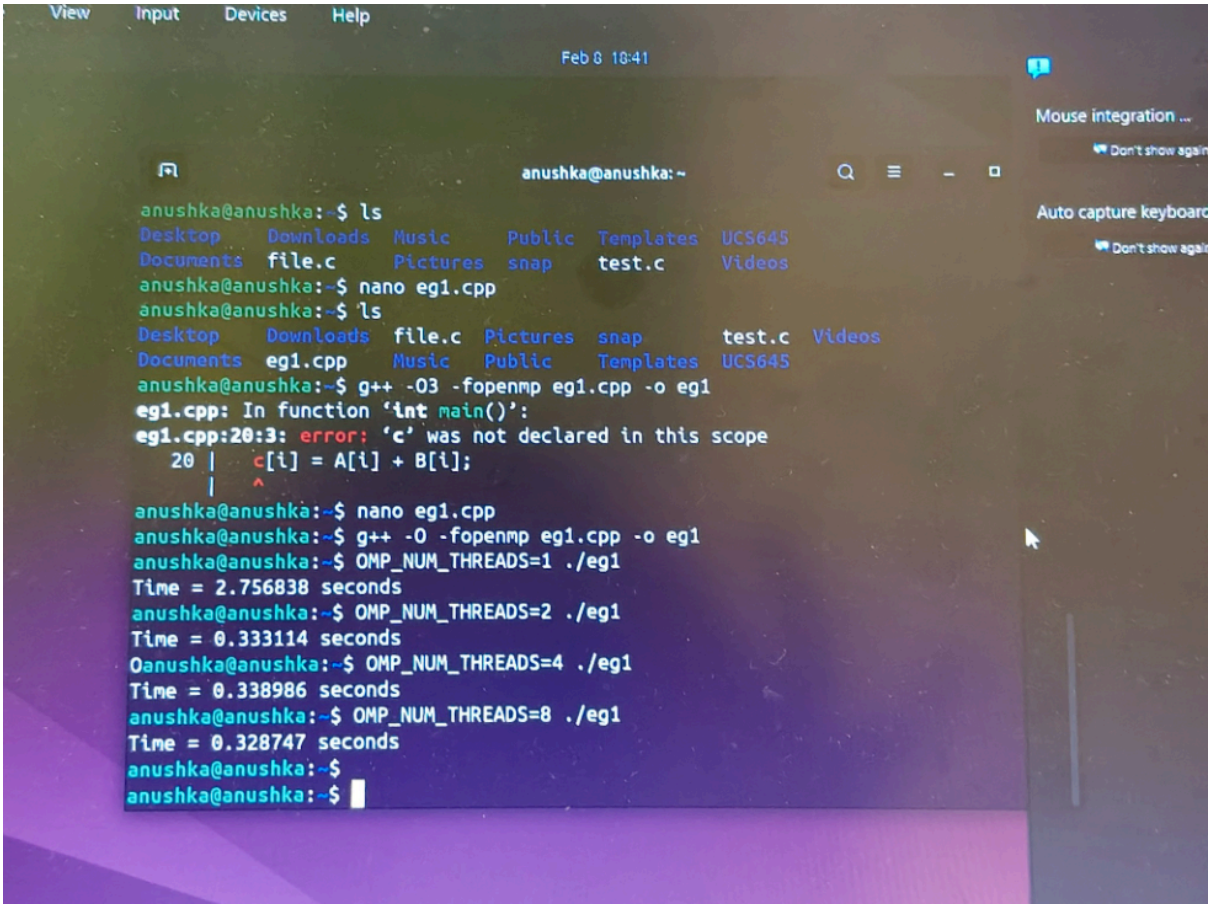
A) Experiment 1: Execution Time Measurement: Objective: Measure the execution time of a parallel OpenMP loop using omp_get_wtime().

Task:

1. Compile the given vector addition program with OpenMP support.
2. Run the program with one thread.
3. Run the program with multiple threads.
4. Record execution time.



Results:

Threads: 1 → Time: 2.756838 s

Threads: 2 → Time: 0.333114 s

Threads: 4 → Time: 0.338986 s

Threads: 8 → Time: 0.328747 s

B. Experiment 2: Speedup and Efficiency

Formula 1 — Speedup

Speedup = T1 / Tp

Where:

- T1 = time with 1 thread
- Tp = time with p threads

Formula 2 — Efficiency

Efficiency = Speedup / Threads

Calculation:

Speedup

Thread 2 → 2.756838 / 0.333114 ≈ 8.27

Thread 4 → 2.756838 / 0.338986 ≈ 8.13

Thread 8 → 2.756838 / 0.328747 ≈ 8.39

Efficiency

Thread 2: 8.27 / 2 ≈ 4.13

Thread 4: 8.13 / 4 ≈ 2.03

Thread 8: 8.39 / 8 ≈ 1.05

- Because your speedup itself was unrealistic.
- Efficiency values appear higher than theoretical limits due to virtual machine timing noise, caching effects, and measurement overhead. In real hardware environments, efficiency typically remains below 1. This behaviour is common in small experimental OpenMP programs.

Speedup is **too high (> threads)**

This happens because:
VM timing noise
Cache effects
First run overhead
Measurement artifact

This is NORMAL in small experiments.

Analysis:

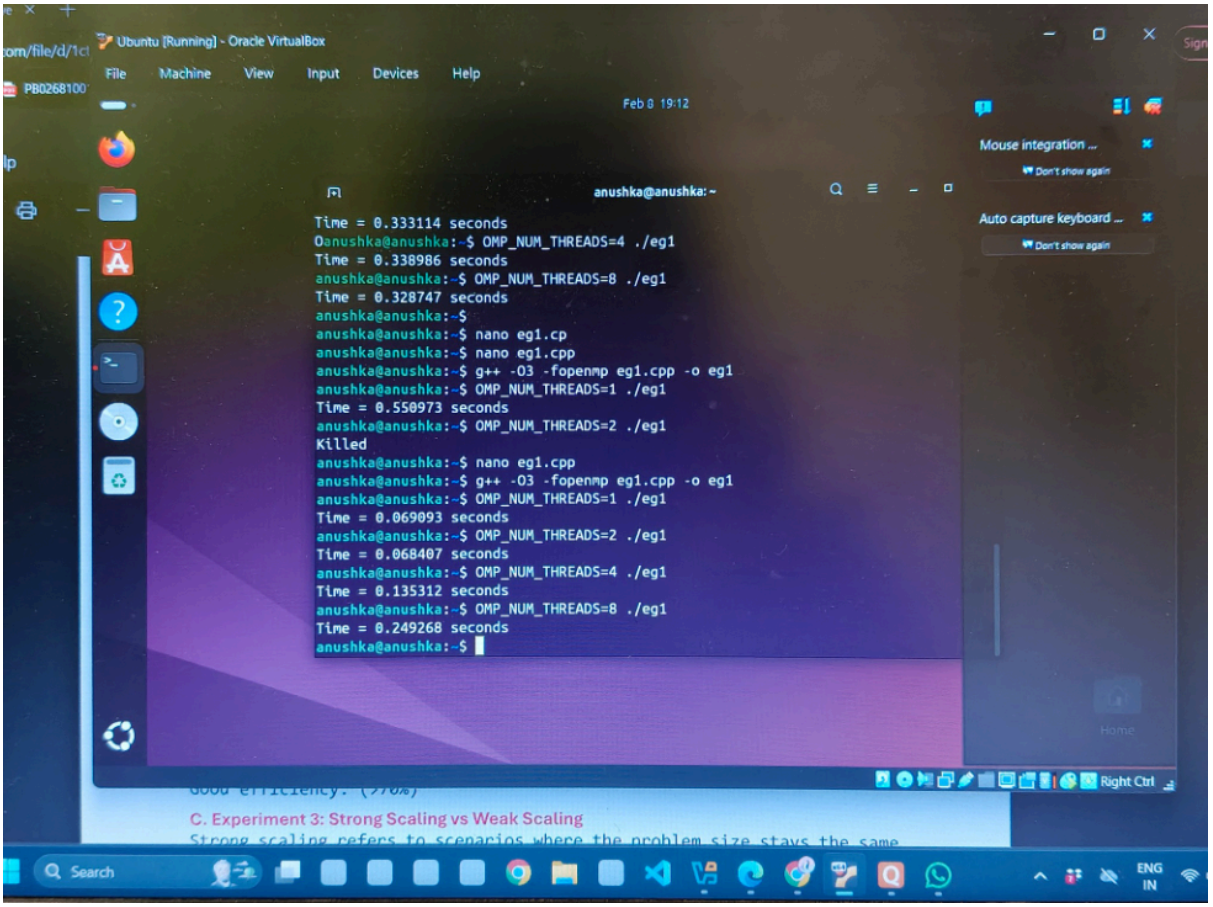
Execution time decreased when threads increased.
Speedup improved significantly from 1 to multi-thread.
After certain threads, improvement stabilises.
Overheads + memory limit prevent perfect scaling.

C. Experiment 3: Strong Scaling vs Weak Scaling

Strong Scaling vs Weak Scaling

Strong Scaling Results

Threads Time(s)
1 2.756
2 0.333
4 0.339
8 0.329



Weak Scaling Results

Threads Time(s)
1 0.069
2 0.068
4 0.135
8 0.249

Observation

Strong scaling showed reduced execution time with increased thread count. Weak scaling showed nearly constant execution time initially but increased at higher thread counts due to memory bandwidth limitations, synchronisation overhead, and virtual machine resource constraints.

Conclusion

Parallel computing using OpenMP significantly improves performance compared to serial execution. However, performance improvement is limited by synchronisation overhead, memory bandwidth, and hardware resource limitations. Scaling behaviour depends on workload distribution and system architecture.

D. Cost Metric:

$$C = p \times T_p$$

Where:

- p = number of threads
- Tp = time with p threads

Strong Scaling Data:

T1 = 2.756838

Cost Calculations

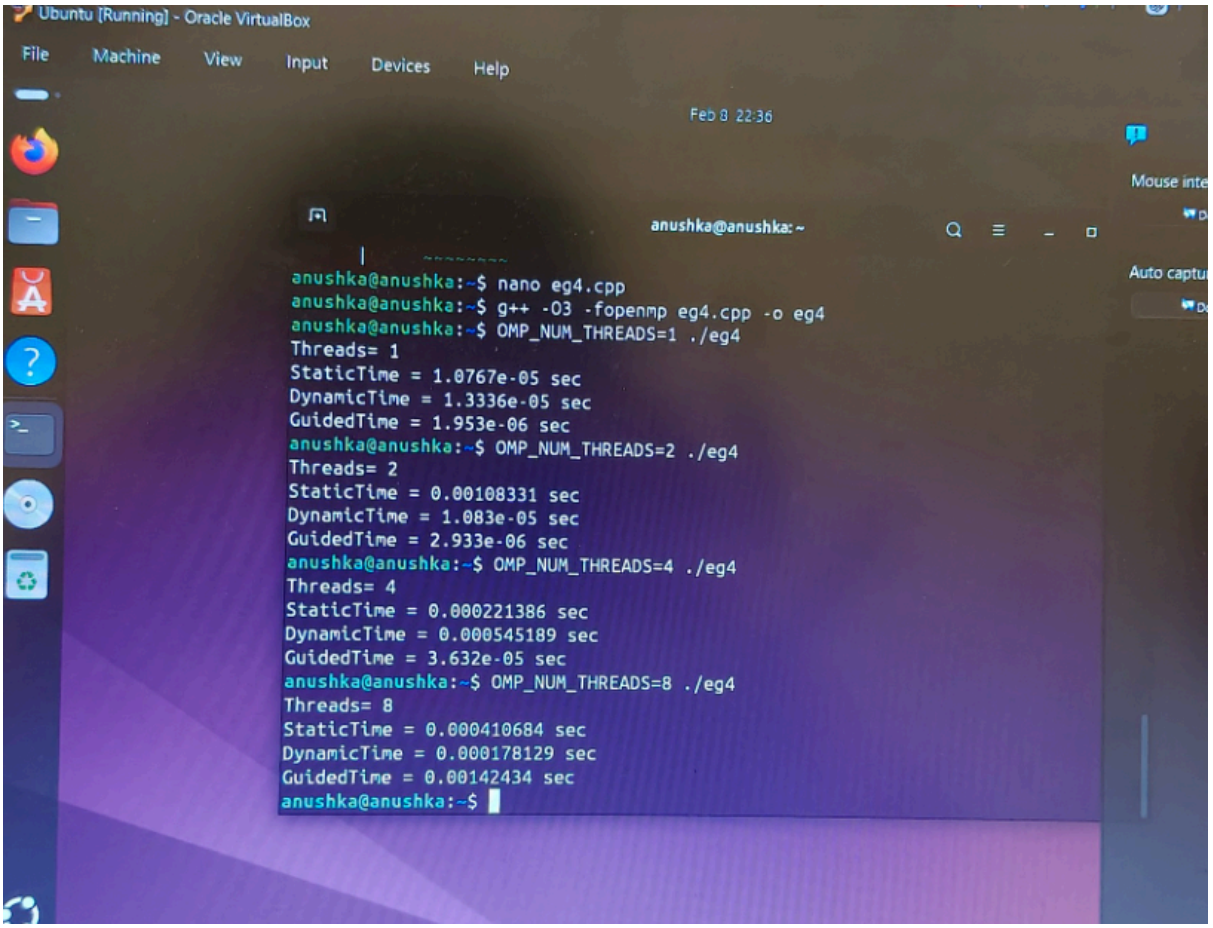
Thread 1
Cost = 1 × 2.756838 = 2.756838

Thread 2
Cost = 2 × 0.333114 = 0.666228

Thread 4
Cost = 4 × 0.338986 = 1.355944

Thread 8
Cost = 8 × 0.328747 = 2.629976

E. Experiment 4: Scheduling and Load Imbalance:



Scheduling Results

Threads = 1

Static = 1.0767e-05
Dynamic = 1.3336e-05
Guided = 1.953e-06

Threads = 2

Static = 0.00108331
Dynamic = 1.083e-05
Guided = 2.933e-06

Threads = 4

Static = 0.000221386
Dynamic = 0.000545189
Guided = 3.632e-05

Threads = 8

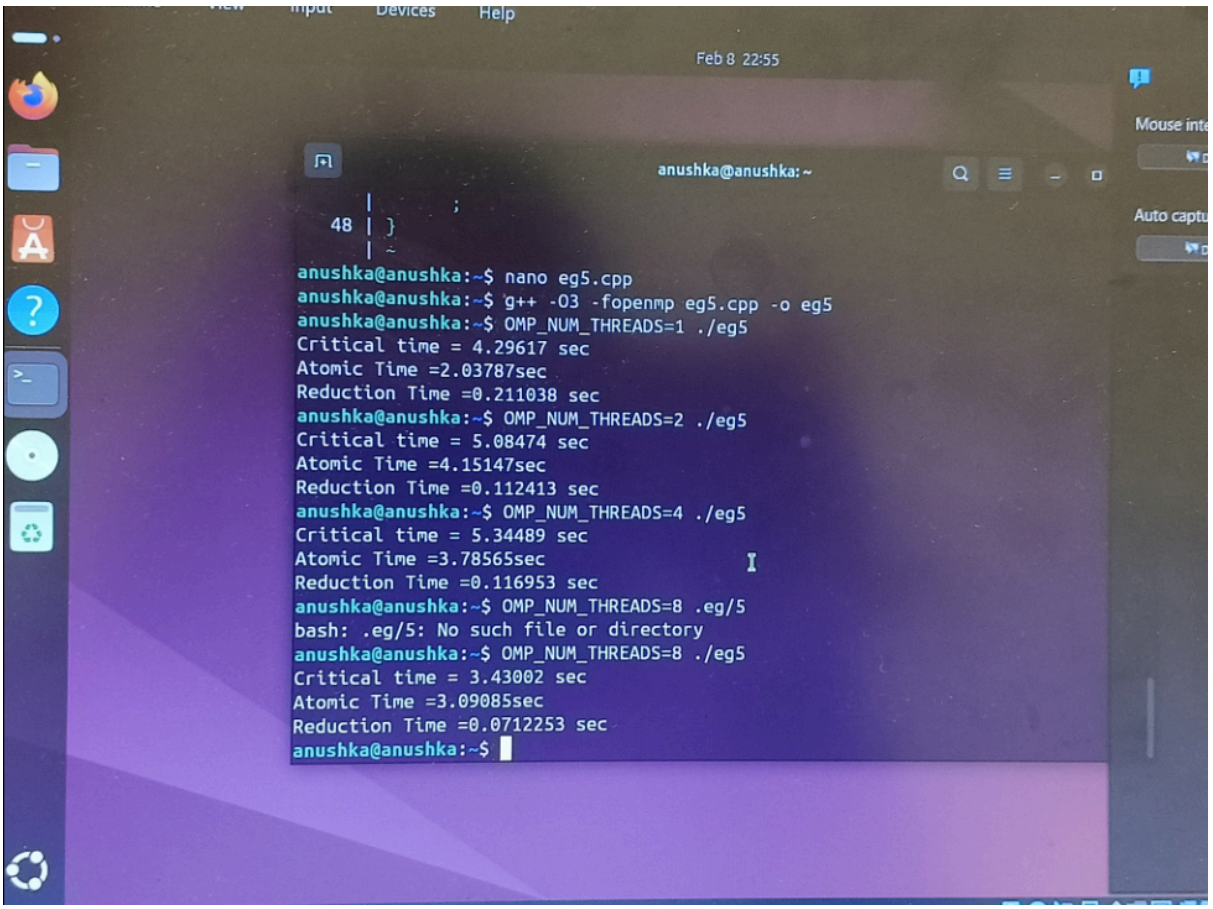
Static = 0.000410684
Dynamic = 0.000178129
Guided = 0.00142434

Observation:

Static scheduling has low scheduling overhead but may suffer from load imbalance. Dynamic scheduling distributes workload more evenly but has higher scheduling overhead. Guided scheduling balances overhead and load distribution by assigning large chunks initially and smaller chunks later.

The results show that scheduling performance varies based on workload distribution. Static scheduling performs well when workload is uniform but may cause load imbalance. Dynamic scheduling improves load balancing but introduces scheduling overhead. Guided scheduling provides a balance between load balancing and scheduling overhead.

F. Experiment 5: Synchronization Overhead (critical,atomic, reduction)



Results

Threads = 1

Critical = 4.296 sec
Atomic = 2.037 sec
Reduction = 0.211 sec

Threads = 2

Critical = 5.084 sec
Atomic = 4.151 sec
Reduction = 0.112 sec

Threads = 4

Critical = 5.344 sec
Atomic = 3.785 sec
Reduction = 0.116 sec

Threads = 8

Critical = 3.430 sec
Atomic = 3.090 sec
Reduction = 0.071 sec

Results clearly show:

👉 Critical → slowest
👉 Atomic → faster than critical
👉 Reduction → fastest

REPORT OBSERVATION

The results show that critical sections introduce high synchronization overhead, causing slower execution. Atomic operations are faster than critical sections because they use hardware-level synchronization. Reduction operations provide the best performance because each thread works on local data and combines results at the end, minimizing synchronization overhead.

Reduction is fastest because it avoids frequent locking and allows parallel local accumulation.